

Stuyvesant Faculty Perspectives:
Mrs. Deena Avigdor:
Seeing the Beauty of Mathematics and New York City

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Mrs. Deena Avigdor has guided students through mathematics at Stuyvesant since 1996. She earned her Bachelor's degree in mathematics from the University of California, Los Angeles, and a Master's degree in Science and Teaching from the University of New Hampshire at the start of her teaching career. Over the past 25 years at Stuyvesant, she has taught the full spectrum of math classes, including Geometry, Precalculus, Calculus Applications, AP Calculus AB, AP Calculus BC, Multivariable Calculus, and Linear Algebra. Currently, she primarily teaches Geometry and AP Calculus BC.

This article is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

1 Learning Math, Teaching Math

From pre-med to math at UCLA—how did this happen? The simple act of sitting in her room reading her Anton Calculus textbook led to Deena Avigdor's road to pure mathematics. Riemann

sums in particular caught her attention.

The abstraction in all forms of mathematics, from Geometry to Algebra, intrigued Avigdor from the beginning. In her opinion, the most fascinating aspect of math is that “you could take something from the physical world and abstract it, and it also works in the reverse. You could take the prime number theorem, something this abstract, and then find applications for it. It kind of works both ways.”

Teaching cemented her career in math. At the parochial high school where Avigdor had attended, her former Calculus teacher broke his hip and a Geometry teacher took maternity leave. These events allowed Avigdor to teach in their place and to realize that math and teaching were her callings.

2 Teaching in Stuyvesant Then

Over the past quarter-century, Avigdor has taught a wide variety of classes, from Geometry to Linear Algebra. When Avigdor first started teaching Geometry, the first math course taken by a typical Stuyvesant student, a former student viewed it as a “downgrade” in course levels for her. However, Avigdor believes that teaching Geometry is just as challenging and rewarding as teaching Calculus, so she thoroughly enjoys teaching both.

Some may have thought math teachers have significantly less work than those in other departments since “[they] don’t have to grade essays.” However, Avigdor explains that “the challenge of dealing with homework in a meaningful way on a daily basis is on par with what other people are doing.” She has found that grading a math test is quite challenging because she has to precisely determine where students went wrong to accurately give them partial credit. There is a significant “gray area” in partial credit as there are various types of mistakes. Avigdor has always endeavored to provide meaningful feedback on every assignment and test for over 150 students, which is quite a feat.

Avigdor recognizes that another challenging aspect of teaching math has been emphasizing and fostering the creativity required to solve complex problems. Although students studying Geometry have to memorize theorems and definitions, these serve as the tools necessary to construct a unique solution. The drawback of high school math is that the curriculum and material are tightly organized in a particular order, but not in the way the material was originally developed. Through proofs and problems with multiple solutions, Avigdor has helped students see the creative aspect of how mathematical tenets came to be.

Avigdor appreciates the unique drive of Stuyvesant students to learn and values how her colleagues provide an environment that has pushed her to grow, whether in math or pedagogy. Three motivators for Avigdor when teaching have been the interaction with the students, the fresh start

every school year, and the evolving curriculum. Furthermore, Avigdor has enjoyed teaching students as freshmen and seeing their growth again as seniors. She remarks, “Sometimes it’s almost not like the same student!” She has also loved the daily joys of a student asking an insightful question or figuring out another solution to a problem. According to Avigdor, introducing proofs has been one of the most challenging parts of teaching because “you are trying to enable the other person to be able to make the connection without you giving it to them.”

3 Teaching at Stuyvesant Now

Even though students are as interested in learning as ever, Avigdor finds that many aspects of teaching at Stuyvesant have changed over time, including the curriculum. For example, much more technology is used now, like Smartboards. In addition, the student population has grown, but the number of teachers has mostly remained the same. Seeing classes grow from 25 students to 34 means having “another row in the back of the classroom.” While a larger number of students means a larger pool of ideas, it also means that teachers cannot give enough individual attention to each person. Avigdor believes a class of 22 to 25 students is most optimal.

When the COVID-19 pandemic prompted remote learning, students collaborated through breakout rooms which were very beneficial. However, the virtual setting still lacked the “buzz” or the “certain energy from being in the classroom around other groups” which normally increases productivity. This year, Avigdor used many platforms such as Jamboard, Google Slides, Classkick, and Desmos, which helped her give feedback to the students. She hopes to integrate the technologies and techniques she’s learned into her in-person classes in the future. Despite her adaptation to remote instruction, she still found this year to be the most challenging of her career, even surpassing the school year of the 9/11 attacks.

Remote learning also introduced varied interaction between students. For example, Avigdor noticed that when the freshmen this year were put in groups, they interacted with each other by typing instead of speaking. This was not the case as much with her senior classes because they knew each other already and had previous experience doing group work in Stuyvesant classes. In addition, Avigdor opted to type comments instead of visiting individual breakout rooms. In the years before the pandemic, she provided directed notes for her Geometry class since it was unnecessary for them to copy definitions off of the board. They instead only had to fill in the blanks in the directed notes. Her Calculus students did not get directed notes, however. For the freshmen this year, she had a “flipped classroom”: Avigdor made videos that they watched for homework to learn the new material, and in class, they solved problems. On the other hand, seniors this year received directed notes since they needed to cover more material in less time than in a typical school year.

4 Beyond the Classroom

So what is the key to a successful career in math? Avigdor advises that it is perseverance. When students train for standardized tests, they tackle short problems and receive immediate answers. However, it takes much longer to find the solution in a mathematical or problem-solving field, so students must learn to stick with it, which is easier said than done.

Impressively, Avigdor has another passion in addition to math: photography. Her interest in photography began back in seventh or eighth grade, and she now immensely enjoys creating photo books with her own layouts and pictures. Interestingly, it was Stuyvesant students enrolled in school photography classes who encouraged her to invest in a sophisticated camera. This summer, Avigdor is taking two photography classes while she continues to work on a project she started about eight years ago. She is determined to capture the beauty of New York City by photographing flowers, which isn't typically associated with New York City, in the Brooklyn Botanical Garden and locally in her neighborhood. Given Avigdor's enthusiastic and dedicated attitude, she will be able to add this project to her photo book library in the near future.